# Structures in deep neural networks

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# Outline

**Outline of the talk**

I. Deep learning and deep neural networks

II. Classical mathematical theory of neural networks

III. Deep ReLU networks in deep learning

IV. Mathematical theory of deep CNNs

V. Superiority of deep CNNs

VI. Transformers for distribution regression

## I. Deep learning and deep neural networks

### I.1. Applications of deep learning

**Big data** leads to scientific challenges: storage bottleneck, algorithmic scalability, ...

**Applications of deep learning**

**Speech Recognition**: from hidden Markov models and Gaussian mixture models to restricted Boltzmann machines and end-to-end deep learning speech recognition systems, phoneme error rate brought down from 26% to 17.7%

**Computer Vision**: Google Street View, autonomous driving, ...

**Natural Language Processing**: ChatGPT, ...

## I.2. Deep neural network architectures

**Convolutional neural networks** (CNNs) introduced for speech and image applications:
Waibel-Hanazawa-Hinton-Shikano-Lang, Zhang-Itoh-Tanida-Ichioka, LeCun-Bottou-Bengio-Haffner, Hinton-Osindero-Teh, Bengio, LeCun, Krizhevsky-Sutskever-Hinton, Simonyan-Zisserman, Mallat, ...

**Recurrent neural networks**: Szegedy-Liu-Jia-Sermanet-Reed-Anguelov-Erhan-Vanhoucke-Rabinovich, ...

Connections to **generative adversarial networks** (GANs)

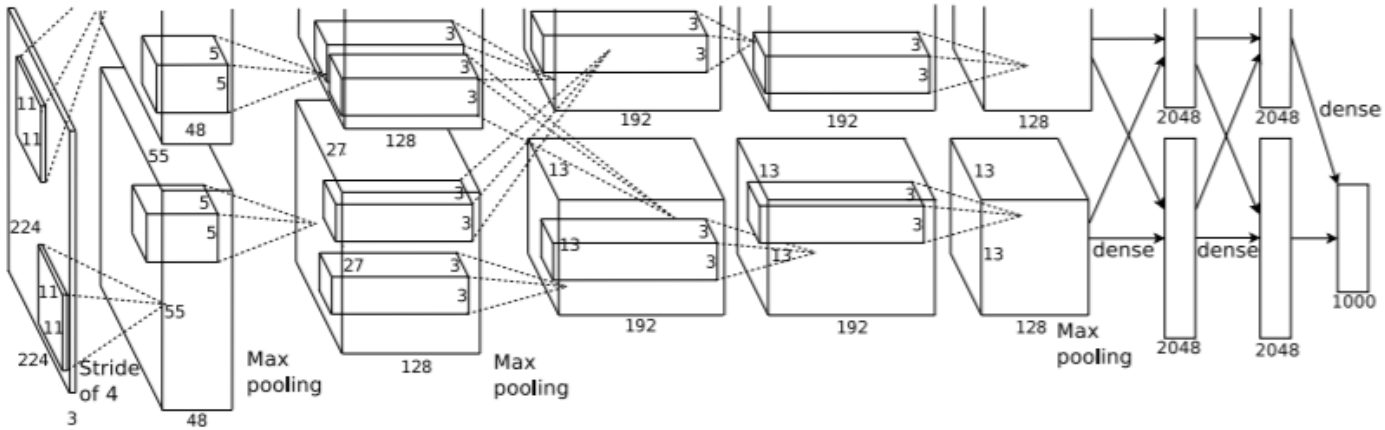
**Attention-based networks**: Transformers, diffusion models, LLMs, ...

## I.3. **Example: AlexNet** (Krizhevsky-Sutskever-Hinton, 2012)

5 layers of CNNs with $\{0,1,\ldots,s\}^2$ matrix filters with $s = 10, 4, 2, 2, 2$ and 3 layers of fully-connected nets with polling and normal layers

convolution of 2D sequences:

$$(W*X)_\alpha = \sum_\beta W_\beta X_{\alpha-\beta} = \left\langle (W_\beta)_{\beta\in\{0,\ldots,s\}^2}, (X_{\alpha-\beta})_{\beta\in\{0,\ldots,s\}^2} \right\rangle_{\ell^2}$$



## II. Classical mathematical theory of neural networks

### II.1. Hypothesis space of fully connected networks

Classical shallow fully connected neural networks on $\mathbb{R}^d$:

$$f_N(x) = f_{T^{(1)},b,c}(x) = \sum_{i=1}^{N} c_i \sigma(\mathbf{w}_i \cdot x + b_i) = c^T \sigma(T^{(1)} x + b),$$

where $x \in \mathbb{R}^d$, activation $\sigma : \mathbb{R} \to \mathbb{R}$ acts componentwise

$$T^{(1)} = [\mathbf{w}_1, \ldots, \mathbf{w}_N]^T = \begin{bmatrix} \mathbf{w}_1^T \\ \vdots \\ \mathbf{w}_N^T \end{bmatrix} \in \mathbb{R}^{N \times d}, \; b, \; c \in \mathbb{R}^N$$

Activation function in the classical literature: sigmoid, tanh, ...

# Classical fully connected networks

Multi-layer neural networks with $J$ hidden layers of neurons $\{h^{(j)} : \mathbb{R}^d \to \mathbb{R}^{d_j}\}_{j=1}^J$ with widths $\{d_j \in \mathbb{N}\}$ and $h^{(0)}(x) = x$

$$h^{(j)}(x) = \sigma\left(T^{(j)} h^{(j-1)}(x) + b^{(j)}\right), \qquad j = 1, \ldots, J \tag{1}$$

**full** connection matrices $T^{(j)} \in \mathbb{R}^{d_j \times d_{j-1}}$ and biases $b^{(j)} \in \mathbb{R}^{d_j}$

Hypothesis space $\mathcal{H} :=$
$\left\{ f_{\mathbf{T},\mathbf{b},c}(x) = c^T h^{(J)}(x) : \mathbf{T} = (T^{(j)})_{j=1}^J, \mathbf{b} = (b^{(j)})_{j=1}^J, c \in \mathbb{R}^{d_J} \right\}$

**Key challenge** for implementation with big data:

The total number of free parameters to train, $\mathcal{N} = \sum_{j=1}^J \left(d_j d_{j-1} + d_j\right) + d_J$, increases very fast when the data dimension $d = d_0$ or the depth $J$ is large

# Structured neural networks

**II.2. Deep learning**: imposing structures to reduce the number of parameters and computational complexity of neural networks

Large literature on local or global convergence of Stochastic Gradient Descent for learning

$$f_{\mathbf{z}} = \arg\min_{\mathbf{T},\mathbf{b},c} \mathcal{E}_{\mathbf{z}}(f_{\mathbf{T},\mathbf{b},c}) := \frac{1}{m}\sum_{i=1}^{m}(f_{\mathbf{T},\mathbf{b},c}(x_i) - y_i)^2$$

Allen-Zhu-Li-Liang, Arora-Du-Hu-Li-Wang, Cao-Gu, Lei-Ying, Lei-Jin-Ying, Lei-Wang-Ying-Zhou, ...

Sample error estimates with norm bounds for $\mathbf{T}$, $\mathbf{b}$, $c$:

Kohler-Krzyżak, Schmidt-Hieber, ...

## II.3. Approximation theory of fully connected networks

**Universality**: If $\sigma$ is locally bounded, piecewise continuous, and not a polynomial, then for any compact $\Omega \subset \mathbb{R}^d$ and $f \in C(\Omega)$, there holds

$$\lim_{N\to\infty} \inf \left\{ \|f - f_{T^{(1)},b^{(1)},c}\|_{C(\Omega)} : \ T^{(1)} \in \mathbb{R}^{N\times d}, b^{(1)}, c \in \mathbb{R}^N \right\} = 0.$$

**Classical theory**: Cybenko (1989), Hornik (1991), Barron (1993), Mhaskar (1993), Mhaskar-Micchelli (1994), Chui-Li-Mhaskar, Lin-Pinkus-Schocken, Maiorov, Petrushev, ...

Typical result (Mhaskar 1993): Assume a continuous activation functions $\sigma$ satisfies with some $b \in \mathbb{R}$, $\sigma^{(k)}(b) \neq 0$ for $k \in \mathbb{Z}_+$ and with some integer $\ell \in \mathbb{N}$, $\lim_{u\to-\infty} \sigma(u)/|u|^\ell = 0$ and $\lim_{u\to\infty} \sigma(u)/u^\ell = 1$. Then for $f \in C^r([-1,1]^d)$,

$$\inf \left\{ \|f - f_N\|_{C(\Omega)} : \ \mathbf{w}_i, b, c \right\} = O(N^{-r/d}) = O\left( (d+2)^{r/d} \mathcal{N}^{-r/d} \right).$$

## III. Deep ReLU networks in deep learning

### III.1. Rectified linear unit (ReLU)

Most commonly used activation function in deep learning: Rectified linear unit (**ReLU**) given by $\sigma(u) = (u)_+ = \max\{u, 0\}$
**Advantages** of the induced networks: simple implementation, overcome the difficulty of gradient vanishing in training

**Main problem**: ReLU is not $C^\infty$ and the classical condition fails

Telgarsky (2016), Yarotsky (2017): approximation by deep fully connected ReLU networks
Bach, Petersen, Voigtlaender, Bölcskei, Grohs, Kutyniok, Mhaskar-Poggio, Suzuki, E-Ma-Wu, Lu-Shen-Yang-Zhang, Montanelli-Du, He-Li-Xu, Chui-Lin-Zhou, ...
Generalization analysis: Schmidt-Hieber (2020), Kohler and Langer (2021), Liu-Jiao-Wang-Huang (2025), ...

### III.2. Shallow nets are enough for realizing smoothness

Recently, it is found that **shallow ReLU networks**

$$f_N(x) = \sum_{i=1}^{N} c_i \sigma(\mathbf{w}_i \cdot x + b_i)$$

are similar to sigmoid networks and **enough for realizing smoothness in regression**:

Klusowski-Barron (2018) for functions in the Barron space:
$\|f - f_N\|_\infty = O(N^{-\frac{1}{2}})$ if $\int |\omega|^2 |\hat{f}(\omega)| < \infty$

Siegel-Xu (2020), Siegel (2023): if $f(x) = \int_{\mathbb{S}^d} \sigma\left((x,1)\cdot v\right) d\mu(v)$

Mao-Zhou (2023): $\|f - f_N\|_\infty = O((\log N)^{\frac{1}{2}+d} N^{-\frac{r}{d}\frac{d+2}{d+4}})$ if
$f \in W^r_\infty$, $r < \frac{d}{2} + 2$

# Theory of ReLU networks

## Theorem 1

*Let* $f \in W_\infty^r(\{x \in \mathbb{R}^d : \|x\| \le 1\})$. *If* $0 < r < (d+3)/2$, *then* $\|f - f_N\|_\infty = O(N^{-\frac{r}{d}})$. *If* $r > (d+3)/2$, *then* $\|f - f_N\|_\infty = O(N^{-\frac{d+3}{2d}})$.

Minimax rates of convergence for learning a regression function $f_\rho \in W_\infty^r(\{x \in \mathbb{R}^d : \|x\| \le 1\})$ by least squares regression with shallow ReLU networks:

$O(m^{-\frac{2r}{2r+d}} \log m)$ if $r < (d+3)/2$ and $O(m^{-\frac{d+3}{2d+3}} \log m)$ if $r > (d+3)/2$, where $m$ is the sample size

[1] Y. F. Yang and D. X. Zhou, Nonparametric regression using over-parameterized shallow ReLU neural networks, Journal of Machine Learning Research **25**(165):1–35, 2024.

### III.3. Questions on roles of network structures

Shallow ReLU networks are enough to capture Hölder smoothness of functions. But deep neural networks, often with structures, are powerful in practice.

**Desired mathematical theory for deep learning**: Why and how do the **structures**of deep neural networks improve the performance in **learning functions with structures**?

**Why are CNNs powerful in processing natural images and speeches?** It is believed that CNNs extract local shift invariance of natural images and speeches.

**Why are transformers powerful in natural language processing?** It is believed that transformers extract sequential dependence of natural language data.

## IV. Mathematical theory of deep CNNs

### IV.1. Convolutions and convolutional matrices

The convolution of two sequences $w = (w_k)_{k=-\infty}^{\infty}$ and $x = (x_k)_{k=-\infty}^{\infty}$ is another sequence $w*x$ given by $(w*x)_i = \sum_{k=-\infty}^{\infty} w_{i-k} x_k$ for $i \in \mathbb{Z}$.

If $w$ is supported in $\{0, 1, \ldots, s\}$ and $x$ is supported in $\{1, \ldots, d\}$, then $w*x$ is supported in $\{1, \ldots, d+s\}$ and

$$(w*x)_i = \sum_{k=0}^{s} w_k x_{i-k} = w_s x_{i-s} + w_{s-1} x_{i-s+1} + \ldots + w_0 x_i.$$

**Zero-padding**: if $x$ is unknown or unavailable outside $\{1, \ldots, d\}$ we take zero terms outside. This leads to the input-output relation with a matrix $T^w$:

$$[(w*x)_i]_{i=1}^{d+s} = T^w [x_k]_{k=1}^{d}$$

# convolutional matrices

**Convolutional matrix** with zero-padding:

$T^w = (w_{i-k})_{1\le i\le d+s, 1\le k\le d}$

$$\begin{bmatrix} (w*x)_1 \\ (w*x)_2 \\ \vdots \\ (w*x)_{i-1} \\ (\mathbf{w}*\mathbf{x})_{\mathbf{i}} \\ (w*x)_{i+1} \\ \vdots \\ (w*x)_d \\ (w*x)_{d+1} \\ \vdots \\ (w*x)_{d+s} \end{bmatrix} = \begin{bmatrix} w_0 & 0 & 0 & \cdots & \cdots & 0 \\ w_1 & w_0 & 0 & \cdots & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ w_s & \cdots & \cdots & w_0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & w_s & \cdots & w_1 & w_0 \\ 0 & \cdots & 0 & w_s & \cdots & w_1 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & w_s \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{i-s-1} \\ \mathbf{x}_{\mathbf{i}-\mathbf{s}} \\ \vdots \\ \mathbf{x}_{\mathbf{i}} \\ x_{i+1} \\ \vdots \\ x_d \end{bmatrix}.$$

## IV.2. Deep CNNs induced by convolutions

Waibel-Hanazawa-Hinton-Shikano-Lang: time-delay neural networks (1989), Zhang-Itoh-Tanida-Ichioka: shift invariant neural networks (1990)

Convolutional filters $\mathbf{w} = \{w^{(j)} = (w_k^{(j)})_{k=-\infty}^{\infty}\}_{j=1}^{J}$

**filter length** $s$: $w^{(j)}$ is supported in $\{0, 1, \ldots, s\}$

A **deep CNN of depth** $J$ with increasing widths $\{d_j = d + js\}$ and a convolutional sparse structure:

$$h^{(j)}(x) = \sigma\left(T^{(j)} h^{(j-1)}(x) - b^{(j)}\right), \qquad j = 1, 2, \ldots, J, \quad (2)$$

where $T^{(j)} = T^{w^{(j)}}$ is a $d_j \times d_{j-1}$ convolutional matrix generated by the filter $w = w^{(j)}$.

# advantages and challenges

The generated hypothesis space of functions:

$$\mathcal{H} = \cup_{\mathbf{w},\mathbf{b}} \mathcal{H}_J^{\mathbf{w},\mathbf{b}} := \left\{ \sum_{k=1}^{d+Js} c_k h_k^{(J)}(x) : c \in \mathbb{R}^{d+Js} \right\} \tag{3}$$

Free parameters: filters $\mathbf{w} = (w^{(j)})_{j=1}^J$, biases $\mathbf{b} = \{b^{(j)}\}_{j=1}^J$ of the form $b = [b_1 \ \ldots b_{s-1} \ b_s \ b_s \ \ldots \ b_s \ b_{d_j-s+1} \ \ldots b_{d_j}]^T$

**Computational advantage**: the number of free parameters: $\mathcal{N} = 3s(J-1) + 2d + 2J$ with depth $J$, filter length $s$, and data dimension $d$; flexibility in choosing the filters; benign overfitting

**Challenges**: the hypothesis space is the union of infinitely many RKHSs $\Longrightarrow$ lack of robustness or explainability; difficulty in understanding the modeling or generalization ability

## IV.3. Universality of deep CNNs

**Theorem 2**

*Let $2 \le s \le d$. For any compact subset $\Omega$ of $\mathbb{R}^d$ and any $f \in C(\Omega)$, there exist sequences* $\mathbf{w}$ *of filters,* $\mathbf{b}$ *of bias vectors, and $f_J^{\mathbf{w},\mathbf{b}} \in \mathcal{H}_J^{\mathbf{w},\mathbf{b}}$ such that $\lim_{J\to\infty} \|f - f_J^{\mathbf{w},\mathbf{b}}\|_{C(\Omega)} = 0$.*

[2] D. X. Zhou, Universality of deep convolutional neural networks, Appl. Comput. Harmonic Anal. **48** (2020), 787-794.

Key observation: If $W = (W_k)_{k=-\infty}^{\infty}$ is supported in $\{0, \ldots, \mathcal{M}\}$, then there exist filters $\{w^{(j)}\}_{j=1}^{p}$ each supported in $\{0, \ldots, s\}$ with $p \le \lceil \frac{\mathcal{M}}{s-1} \rceil$ satisfying $W = w^{(p)} * w^{(p-1)} * \ldots * w^{(2)} * w^{(1)}$.

[3] S. B. Lin, K. D. Wang, Y. Wang, and D. X. Zhou, Universal consistency of deep convolutional neural networks, IEEE Trans. Inform. Theory **68** (2022), 4610–4617.

## IV.4. Rates of approximation

Sobolev space $H^r(\mathbb{R}^d)$: $F$ and all its partial derivatives up to order $r$ are square integrable on $\mathbb{R}^d$.

Embedding Theorem: $H^r(\mathbb{R}^d) \subset C(\mathbb{R}^d)$ only when $r > \frac{d}{2}$.

### Theorem 3

*If $\Omega \subseteq [-1,1]^d$ and $f$ is the restriction to $\Omega$ of some function $F \in H^r(\mathbb{R}^d)$ on $\mathbb{R}^d$ with integer $r > 2 + \frac{d}{2}$ and $J \geq 2d/(s-1)$, then there exist $\mathbf{w}$, $\mathbf{b}$, and $f_J^{\mathbf{w},\mathbf{b}} \in \mathcal{H}_J^{\mathbf{w},\mathbf{b}}$ such that*

$$\|f - f_J^{\mathbf{w},\mathbf{b}}\|_{C(\Omega)} \leq \widetilde{c}\, \|F\|_{H^r} \sqrt{\log J}\, (1/J)^{\frac{1}{2}+\frac{1}{d}},$$

*where $\widetilde{c}$ is an absolute constant and $\|F\|_{H^r}$ denotes the Sobolev space norm of $F$.*

$\|f - f_J^{\mathbf{w},\mathbf{b}}\|_{C(\Omega)} = O\left(1/\sqrt{J}\right)$ where $J$ is the depth

# Superiority of CNNs

## V. Superiority of deep CNNs

**CNNs are at least as good as fully connected nets**: The output of any fully connected ReLU network can be realized by a (downsampled) deep CNN with at most 8 times parameter number.

[4] D. X. Zhou, Theory of deep convolutional neural networks: Downsampling, Neural Networks **124** (2020), 319-327.

**Superiority in approximating ridge functions**: $g(\xi \cdot x)$ with unknown $\xi \in \mathbb{R}^d$ and Lipschitz-$\alpha$ ($0 < \alpha \leq 1$) univariate $g$ can be approximated by deep CNNs, as for approximating the univariate $g$ by fully connected networks.

[5] Z. Y. Fang, H. Feng, S. Huang, and D. X. Zhou, Theory of deep convolutional neural networks II: Spherical analysis, Neural Networks **131** (2020), 154-162.

# Superiority for extracting radial features

**Superiority in approximating radial functions**: $f(|x|^2)$

Consider $\mathbb{B} := \{x \in \mathbb{R}^d : |x| \le 1\}$, the space $C^{0,1}(\mathbb{B})$ of Lipschitz functions with norm $\|\cdot\|_{C^{0,1}(\mathbb{B})}$ and radial function set

$$\mathcal{B}\left(C^{0,1}_{|\cdot|}\right) := \left\{f(|\cdot|^2) : \|f(|\cdot|^2)\|_{C^{0,1}(\mathbb{B})} \le 1\right\}.$$

Denote the span of $N$ ridge functions as

$$\mathcal{S}_N = \left\{\sum_{k=1}^{N} c_k \sigma_k(a_k \cdot x - b_k) : \ \sigma_k \in C(\mathbb{R}),\ a_k \in \mathbb{R}^d,\ c_k, b_k \in \mathbb{R}\right\}.$$

Shallow neural network: $\sigma_1 = \ldots = \sigma_N$

The efficiency of a hypothesis space $V$ in approximating a function set $U$ is measured by the quantity

$$\operatorname{dist}(U, V) := \sup_{f \in U} \inf_{g \in V} \|f - g\|_{L_\infty(\mathbb{B})}.$$

# Learning radial functions

## Theorem 4

*Let* $2 \le s \le d$. *We have* $dist\left(\mathcal{B}\left(C^{0,1}_{|\cdot|}\right), \mathcal{S}_N\right) \ge c_d N^{-\frac{1}{d-1}}$ *with a constant* $c_d$ *independent of* $N \in \mathbb{N}$; *while*

$$dist\left(\mathcal{B}\left(C^{0,1}_{|\cdot|}\right), \mathcal{H}_N\right) \le 3\sqrt{1+4d}N^{-\frac{1}{2}}, \qquad \forall N \in \mathbb{N}$$

*achieved* $\mathcal{H}_N$ *of a deep CNN of depth* $J = \left\lceil \frac{(2N+3)d}{s-1} \right\rceil$ *followed by a fully connected layer of widths* $2N+3$.

Complexity for accuracy $\epsilon > 0$: $O(\epsilon^{-2})$ for the deep CNN net; $O(\epsilon^{-(d-1)})$ for fully connected shallow nets.

[6] T. Mao, Z. J. Shi, and D. X. Zhou, Theory of Deep Convolutional Neural Networks III: Approximating radial functions, Neural Networks **144** (2021), 778–790.

**Learning functions with multi-features**

$$f(x) = g\left(F_1(x), F_2(x), \ldots, F_{d^*}(x)\right), \qquad x \in \mathbb{R}^d,$$

with multiple general features $F_1, F_2, \ldots, F_{d^*}$ and an outer task-oriented function $g$

Hierarchical interaction model; functions with polynomial features; multiple symmetric polynomial features; multiple spatially sparse smooth features

[7] T. Mao, Z. J. Shi, and D. X. Zhou, Approximating functions with multi-features by deep convolutional neural networks, Anal. Appl. **21** (2023), 93–125.

Learning in Korobov spaces

## VI. Transformers for distribution regression

For a compact set $\Omega \subset \mathbb{R}^d$, denote the collection of all Borel probability measures on $\Omega$ by $\mathcal{P}(\Omega)$. It is a compact metric space (Wasserstein space) $\mathcal{U} := (\mathcal{P}(\Omega), W_2)$ with Wasserstein metric $W_2$.

Distribution regression aims at learning a function $f : \mathcal{U} \to \mathbb{R}$.

We model the learning process by a Borel probability measure $\rho$ on $\mathcal{Z} = \mathcal{U} \times \mathcal{Y}$ with $\mathcal{Y} = [-M, M]$ for some $M > 0$.

Then the target regression function of distribution regression is given by means $f_\rho(\mu) = \int_{\mathcal{Y}} y d\rho(y|\mu)$ of the conditional distributions $\rho(\cdot|\mu)$ at $\mu$. It minimizes the mean squared error $\mathcal{E}(f) = \int_{\mathcal{Z}} (f(\mu) - y)^2 d\rho$. But $\rho$ is unavailable.

# Distribution regression

## VI.1. Two-stage sampling in distribution regression

First-stage data $D = \{(\mu_i, y_i)\}_{i=1}^m$ i.i.d. sampled from $\rho$.

But $\mu_i$ is available only through a second-stage i.i.d. sample $\{x_{i,j} \in \Omega\}_{j=1}^{n_i}$. second-stage data:

$$\hat{D} = \left\{ \left( \{x_{i,j}\}_{j=1}^{n_i}, y_i \right) \right\}_{i=1}^m$$

and the available empirical distribution $\hat{\mu}_i^{n_i} = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{x_{i,j}}$.

**Empirical risk minimization** (ERM) over a hypothesis space $\mathcal{H}$ of continuous functions on $\mathcal{P}(\Omega)$:

$$f_{\hat{D},\mathcal{H}} = \arg\min_{f \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \left( f(\hat{\mu}_i^{n_i}) - y_i \right)^2 .$$

Classical approach of kernel mean embedding:
Z. Szabó, B. Sriperumbudur, B. Póczos, and A. Gretton (2016).

## VI.2. Transformers for distribution regression

With an input sequence $Q = [x_1 \cdots x_m]^T \in \mathbb{R}^{m\times d}$ of length $m$ and feature dimension $d$, the single-head attention is defined as

$$\text{SoftmaxAttn}(x_i) = \sum_{j=1}^{m} \frac{\exp\left(\langle W_q x_i, W_k x_j\rangle/\sqrt{d_{in}}\right)}{\sum_{j'=1}^{m}\exp\left(\langle W_q x_i, W_k x_{j'}\rangle/\sqrt{d_{in}}\right)} W_v x_j,$$

where $W_q, W_k \in \mathbb{R}^{d_{in}\times d}$, $W_v \in \mathbb{R}^{d\times d}$ are parameter matrices. We view the above as $\frac{1}{m}\sum_{j=1}^{m} k(x_i, x_j) f(x_j)$ with a data-dependent kernel $k$ on $\Omega\times\Omega$ and $f : \Omega \to \mathbb{R}$ and introduce an attention operator $\text{attn}(\mu) = \int_\Omega k(\cdot, x) f(x) d\mu(x)$ and a transformer encoder as

$$c \cdot \sigma\left(A\left[\text{attn}(\mu)\right]|_{\mathbf{T}} + b\right),$$

where $\mathbf{T}$ is a set of $n_2$ points in $\Omega$, $A \in \mathbb{R}^{n_1\times n_2}$, $b \in \mathbb{R}^{n_1}$ and $c \in \mathbb{R}^{n_1}$.

# Transformers for distribution regression

We generate the hypothesis space by bounding the parameters as $\|A\|_\infty \le R m^{R \log m}$, $\|c\|_1 \le R\sqrt{m}$, $\|b\|_\infty \le R$ with $R > 0$.

Assume a Barron type condition

$$f_\rho = \Phi\left(\int_\Omega k(\cdot, x) f(x) d\mu(x)\right)$$

with a Mercer kernel $k$, $f$ satisfying $c_f \le |f(x)| \le C_f$ with some $c_f > 0$, and the functional $\Phi$ having a representation

$$\Phi(g) = \int_{\mathcal{H}_k} e^{i\langle g,\omega\rangle_k} e^{i\theta(\omega)} F(d\omega), \qquad \|g\|_k \le r$$

with some $r > 0, \theta : \mathcal{H}_k \to \mathbb{R}$ and a nonnegative function $F$ satisfying $\int_{\mathcal{H}_k} \|\omega\|_k F(d\omega) < \infty$. Then the ERM truncated at level $M$ achieves $\mathbb{E}\left[\left\|\pi_M f_{\hat{D},\mathcal{H}} - f_\rho\right\|^2_{L^2_{\rho_\mathcal{U}}}\right] = O\left(\frac{(\log m)^{d+2}}{\sqrt{m}}\right)$.

[8] P. L. Liu and D. X. Zhou, Generalization analysis of transformers in distribution regression, Neural Computation **37** (2025), 260–293.

## VI.3. Linear transformers for in-context learning

For in-context learning we look for a function $\Phi : \mathcal{P}(\Omega) \times \Omega \to \mathbb{R}$ minimizing the generalization error

$$\mathcal{E}(\Phi) = \mathbb{E}_{\rho \sim \mathcal{P}(\Omega)} \mathbb{E}_{X \sim \rho} \left\| \Phi(\rho, X) - Y \right\|^2 .$$

We use linear transformers based on linear attention defined by

$$\sum_{j=1}^{n} \alpha_j \sigma \left( \sum_{q=1}^{k} \phi_{q,m}(x) \left( \sum_{p=1}^{k} \mathcal{T}_C \left[ \int \phi_{p,m}(y) A_{p,q}^{(j)} y d\rho(y) \right] \right) + b_j \right) + b_0,$$

where $A_{p,q}^{(j)} \in \mathbb{R}^{d \times d}$, $b_j \in \mathbb{R}^d$, $\alpha_j \in \mathbb{R}$, $\mathcal{T}_C$ is a truncation operator, and $\phi_{q,m}$ are two-hidden-layer tanh neural networks with product gates. Generalization analysis with two-stage sampling: Liu-Zhou.

**Thank you very much!**